

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) and adhering to the "Coding Requirements" section.

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from a volume and load it into Delta tables for customer and order data.

Preconditions:

- CSV files are available at the specified locations (/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata and /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata)
- Delta tables for customer and order data do not exist or are empty

Main Flow / Functional Steps:

1. Read customer CSV data from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata into a Delta table named "customer".
2. Read order CSV data from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata into a Delta table named "order".
3. Add a new column "TotalAmount" to the "order" Delta table, calculated as the product of "PricePerUnit" and "Qty".

FR-DLT-002: Clean and Process Customer and Order Data

Title: Clean and Process Customer and Order Data

Description: The system must remove null and duplicate records from the customer and order Delta tables.

Preconditions:

- Customer and order Delta tables exist and contain data

Main Flow / Functional Steps:

1. Remove rows with null values from the "customer" and "order" Delta tables.
2. Remove duplicate rows from the "customer" and "order" Delta tables.

FR-DLT-003: Create and Load ordersummary Table

Title: Create and Load ordersummary Table

Description: The system must create an "ordersummary" table if it does not exist and load it with joined customer and order data using SCD type 2.

Preconditions:

- Customer and order Delta tables exist and contain clean data
- "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "ordersummary" table in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist.
2. Join the "customer" and "order" Delta tables on the "CustId" column.
3. Load the joined data into the "ordersummary" table using SCD type 2.

FR-DLT-004: Update ordersummary Table on Customer Changes

Title: Update ordersummary Table on Customer Changes

Description: The system must update the "ordersummary" table when there are changes to the customer data.

Preconditions:

- "ordersummary" table exists and contains data
- Customer data has changed

Main Flow / Functional Steps:

1. Identify changes to the customer data.
2. Update the "ordersummary" table by making old records inactive and new records active.
3. Update the "StartDate" and "EndDate" columns accordingly to maintain history.

FR-DLT-005: Create and Load customeraggregatespend Table

Title: Create and Load customeraggregatespend Table

Description: The system must create a "customeraggregatespend" table and load it with aggregated data from the "ordersummary" table.

Preconditions:

- "ordersummary" table exists and contains data

- "customeraggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "customeraggregatespend" table in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist.
2. Aggregate the "TotalAmount" column from the "ordersummary" table by "Name" and "Date".
3. Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-006: Implement Delta Live Tables**Title: Implement Delta Live Tables**

Description: The system must implement the above requirements using Delta Live Tables.

Preconditions:

- All previous requirements have been implemented

Main Flow / Functional Steps:

1. Implement the above requirements using Delta Live Tables to ensure data is processed and loaded correctly.